

CHAPTER 2: QUANTUM MECHANICAL MODEL OF ATOM

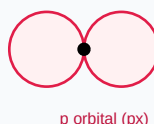
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 1
Chapter Name:	Quantum Mechanical Model of Atom

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



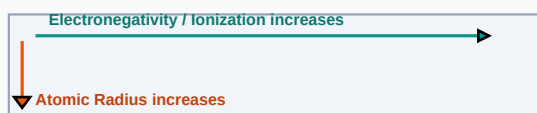
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Quantum Numbers	A set of four integers (n, l, m, s) that describe the energy, shape, orientation, and spin of an electron.
Heisenberg Uncertainty Principle	It is impossible to determine simultaneously both the position and momentum of a subatomic particle with absolute precision.
Orbital	A three-dimensional space around the nucleus where the probability of finding an electron is maximum.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

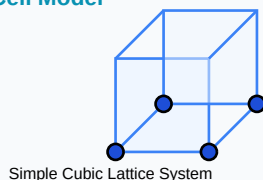
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- De Broglie Wavelength: $\lambda = h / (m \cdot v) = h / p$
- Heisenberg Principle: $\Delta x \cdot \Delta p \geq h / (4 \cdot \pi)$
- Maximum Electrons in Shell: $\text{Max} = 2 \cdot n^2$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

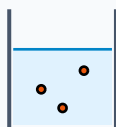
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Calculate de Broglie wavelength of a 0.1kg ball moving at 60 m/s. ($h = 6.626 \times 10^{-34} \text{ J-s}$)

Step-by-step Solution:

$$\lambda = h / (m \cdot v) = 6.626 \times 10^{-34} / (0.1 \times 60) = 6.626 \times 10^{-34} / 6 = 1.1 \times 10^{-34} \text{ meters.}$$

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

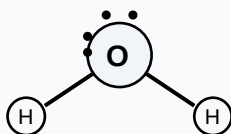
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: State and explain Aufbau Principle, Pauli's Exclusion Principle, and Hund's Rule of Maximum Multiplicity.

Analytical Solution Strategy:

- (1) Aufbau: Orbitals are filled in order of increasing energy ($n+l$ rule).
- (2) Pauli: No two electrons in an atom can have the same set of four quantum numbers.
- (3) Hund: Pairing of electrons in degenerate orbitals does not occur until each orbital is singly occupied.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

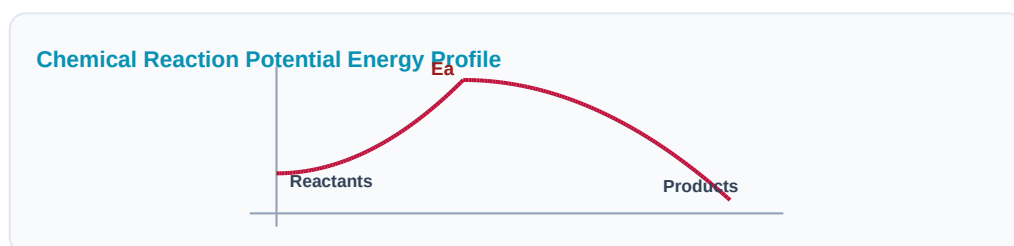
Q1 (1-Mark): Describe the shapes of s, p, and d orbitals.

Q2 (2-Mark): Write down the electronic configuration of Chromium ($Z=24$) and explain its stability anomaly.

Q3 (2-Mark): What are the four quantum numbers? Define each.

Q4 (5-Mark): Calculate the total number of orbitals associated with principal quantum number $n=3$.

Q5 (5-Mark): State Heisenberg's uncertainty principle equation.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Spectroscopy

Laser physics depends on electronic transition jumps between shell states.

Application Case: Materials Engineering

Semiconductor bands are engineered by mapping quantum shells of elements.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

